

# A semi-Lagrangian multi-moment finite volume method with fourth-order WENO projection

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## SUMMARY

Numerical oscillation has been an open problem for high-order numerical methods with increased local degrees of freedom (DOFs). Current strategies mainly follow the limiting projections derived originally for conventional finite volume methods and thus are not able to make full use of the sub-cell information available in the local high-order reconstructions. This paper presents a novel algorithm that introduces a nodal value-based weighted essentially non-oscillatory limiter for constrained interpolation profile/multi-moment finite volume method (CIP/MM FVM) (Li and Xiao, *J. Comput. Phys.*, 222 (2007), 849–871) as an effort to pursue a better suited formulation to implement the limiting projection in schemes with local DOFs. The new scheme, CIP-CSL-WENO4 scheme, extends the CIP/MM FVM method by limiting the slope constraint in the interpolation function using the weighted essentially non-oscillatory (WENO) reconstruction that makes use of the sub-cell information available from the local DOFs and is built from the point values at the solution points within three neighboring cells, thus resulting a more compact WENO stencil. The proposed WENO limiter matches well the original CIP/MM FVM, which leads to a new scheme of high accuracy, algorithmic simplicity, and computational efficiency. We present the numerical results of benchmark tests for both scalar and Euler conservation laws to manifest the fourth-order accuracy and oscillation-suppressing property of the proposed scheme. Copyright © 2016 John Wiley & Sons, Ltd.

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## 1. INTRODUCTION

High-order numerical methods of less numerical diffusion have been getting appealing in solving fluid dynamic problems, particularly in the cases where the flows are vortex dominant. Compared with the conventional finite volume method (FVM), such as the essentially non-oscillatory scheme [1–3] and weighted essentially non-oscillatory (WENO) scheme [4–7], which needs a wide computational stencil to realize high-order accuracy and is difficult to be extended to unstructured meshes, the major merits of high-order schemes with local reconstructions lie in their spectral-like convergence and adaptability to unstructured meshes.

For the past decades in the area of computational fluid dynamics, making use of locally (cell-wisely) increased degrees of freedom (DOFs) has become an active trend to construct high-order schemes. The representative numerical methods of this sort are the spectral element method [8], the discontinuous Galerkin (DG) method [9–13], the staggered-grid Chebyshev multidomain method [14], the spectral volume method [15, 16], and spectral difference method [17]. Recently, a more generalized formulation to reconstruct high-order spectral-convergent schemes,

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the so-called flux reconstruction method, was developed in [18, 19], which can recover nearly all existing nodal value-based high-order schemes. Another more general and intuitive formulation, the multi-moment-constrained flux reconstruction method, was proposed in [20, 21] to provide a more flexible and straightforward way to reconstruct high-order schemes with a wider variety of constraint conditions rather than limited to the correction functions in [18].

An alternative way to increase the local DOFs was introduced in the constrained interpolation profile (CIP) scheme [22, 23], where two types of moments, that is, the point values (PVs) and the first-order derivatives, are simultaneously treated as the model variables and separately updated in time. The successive studies based on CIP scheme result in a series of conservative CIP schemes [24–27], called CIP-conservative semi-Lagrangian (CIP-CSL) schemes, for scalar conservation laws. In CIP-CSL schemes, volume-integrated average (VIA) is treated as a new model variable that is updated in the same manner as the traditional FVM; thus, the scheme has exact numerical conservativeness. CIP-CSL scheme has been used to solve the global shallow-water equations in [28] under the semi-Lagrangian semi-implicit framework (SLSI), among other implementations of CSL schemes in geophysical fluid modeling [29–32]. In the SLSI framework, the advection is solved separately from the geopotential gradient forcing (or equivalently the acoustic effect in gas dynamics). The latter is computed by implicitly solving a Helmholtz equation for geopotential or pressure. To our knowledge, implementation of the semi-Lagrangian schemes to hyperbolic systems with the complete characteristic theory has not been adequately explored.

Based on CIP-CSL scheme, a more generalized FVM framework, the CIP/multi-moment finite volume method (CIP/MM FVM) [33], has been developed and successfully adopted in solving the Euler equations. In the CIP/MM FVM scheme, the moments, both VIA and PV, are updated in a more ‘physically motivated’ and flexible way. The VIA is computed from the FVM with numerical fluxes evaluated on the cell surfaces, while the PV defined at the cell boundary is updated through a semi-Lagrangian approach for the hyperbolic governing equations. In order to solve the Euler equations, the CIP/MM FVM is formulated via the full implementation of characteristic theory in [33].

In spite of the superiority to resolve the complex but smooth flow structures, high-order numerical methods with locally increased DOFs need the nonlinear limiting projection to avoid the oscillations in the presence of discontinuities or large jumps. In the literature, some efforts have been reported to solve the problem, including the total variation diminishing [34]/total variation bounded (TVB) limiter [9–11], WENO limiter [4, 5], and rational function reconstruction [26, 35]. TVB limiter [9–11] has proved its ability in suppressing the numerical oscillations for DG method and some other high-order numerical methods with local reconstructions. Compared with TVB limiter, WENO limiter [4, 5] is more attractive because it is able to suppress the numerical oscillations effectively in the vicinity of discontinuity and retain the highest possible accuracy in smooth region through an automatic weighting switch. The original WENO limiter for DG method [36] stems from the fifth-order FVM WENO scheme, where only the cell average values are used in the WENO reconstruction. This formulation fails to include all available local information (DOFs) in DG method, thus not compatible with the high-order numerical schemes with local reconstructions. Some successive works have attempted to use more local information from the cell-wise high-order polynomial, such as the Hermite WENO limiter for DG method [37, 38]. Similar idea is found in [39] to construct the function-based Hermite WENO limiting for multi-moment finite volume method that uses cell mean values and cell mean derivatives. A more general procedure that converts different moments to a set of mean values and constructs the WENO projection under the conventional FVM framework is reported in [40]. A WENO reconstruction, the so-called simple WENO limiter [41–43], was derived for the DG method, which uses the local DOFs over a compact stencil of only three cells.

We have recently proposed a new scheme, called multi-moment constrained finite volume with WENO limiter of fourth-order (MCV-WENO4) method [44], by incorporating the WENO limiter to the multi-moment constrained FVM of third-order (MCV3) scheme. The new WENO limiter in MCV-WENO4 scheme distinguishes itself from the existing ones by taking fully use of the local nodal values (PV) in the WENO interpolation, which leads to a compact computational stencil involving merely the target cell and its adjacent cells. The MCV-WENO4 scheme shows a new

path towards the better matched implementation of WENO limiter to high-order numerical methods using local DOFs.

In this paper, the nodal value-based WENO limiter has been extended to CIP/MM FVM method to solve the Euler conservation laws by implementing the characteristic theory. The resulting scheme, CIP-CSL with WENO limiter of fourth-order scheme, is based on the CIP-CSL3 scheme [33], while the WENO reconstruction that uses the PVs within three adjacent mesh elements is designed to compute the first-order derivative at the cell center in building up the CIP-CSL3 interpolation function. To our knowledge, there is no research ever reported on adopting the WENO limiter based on sub-cell structures for semi-Lagrangian method to solve the Euler equations.

Numerical results of the presented method for the benchmark tests of both scalar and Euler conservation laws are among those of the best solution quality. Moreover, the present scheme does not need the *ad hoc* TVB criterion to switch the WENO reconstruction, which is the key in other existing schemes to control the excessive numerical dissipation.

This paper is organized as follows. Firstly, the CIP/MM FVM scheme will be reviewed under the formulation of CIP/MM FVM in Section 2, including the implementation to scalar conservation law as well as the Euler equations. In Section 3, we will describe the reconstruction procedure of the nodal value-based WENO limiter, which uses a compact stencil including only the target cell and its immediate neighbors. Section 4 shows the comparison between the Eulerian and semi-Lagrangian schemes based on error measurements, which demonstrates the superiority of the semi-Lagrangian scheme presented in this paper. In Section 5, the numerical results of some widely used benchmark tests are presented to verify the accuracy and robustness of the proposed scheme. Finally, we will give a brief summary to end this paper in Section 6.

## 2. REVIEW OF CONSTRAINED INTERPOLATION PROFILE/MULTI-MOMENT FINITE VOLUME METHOD SCHEME

In this section, we will introduce the numerical formulation of CIP/MM FVM for both scalar conservation law and Euler equations. For Euler equations, we adapt the method proposed in [33], which computes the PV using the semi-Lagrangian solution in terms of the Riemann invariants along characteristic curves, while the VIA is updated in a conservative formulation where the numerical flux is evaluated from the quadrature of PVs at Runge–Kutta substeps.

### 2.1. Scalar conservation law

We firstly consider the following 1D conservative transport equation:

$$\frac{\partial u}{\partial t} + \frac{\partial au}{\partial x} = 0, \quad (1)$$

where  $u(x, t)$  is the solution function, and  $a(x)$  the velocity. The computational domain is divided into  $I$  non-overlapping cells or elements  $\Omega_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ ,  $i = 1, 2, \dots, I$ . Inside each element, we define the following two types of moments, respectively,

- the volume integrated average (VIA) over each cell element

$$\bar{u}_i = \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u(x, t) dx, \quad (2)$$

where  $\Delta x_i = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$ , and

- the PV at the cell boundary

$$u_{i+\frac{1}{2}} = u\left(x_{i+\frac{1}{2}}, t\right). \quad (3)$$

Here, both VIA and PV are simultaneously treated as the model variables, and we can build up the interpolation function inside each element by using both VIA and PV. Several CIP-CSL schemes [24–27] have been proposed in the literature. We introduce the interpolation procedure of the CIP-CSL3 [24] as an example of deriving the CIP-CSL schemes.

**2.1.1. Constrained interpolation profile-conservative semi-Lagrangian scheme with third-order polynomial reconstruction.** In each cell element  $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ , we can construct a cubic interpolation function in a form:

$$\tilde{u}(x) = a_3(x - x_i)^3 + a_2(x - x_i)^2 + a_1(x - x_i) + a_0, \quad (4)$$

where  $x_i$  is the cell central point and the unknown coefficients are computed by imposing the following constraint conditions:

$$\begin{cases} \tilde{u}(x_{i+\frac{1}{2}}) = u_{i+\frac{1}{2}}, \\ \tilde{u}(x_{i-\frac{1}{2}}) = u_{i-\frac{1}{2}}, \\ \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \tilde{u}(x) dx = \bar{u}_i, \\ \left( \frac{d\tilde{u}(x)}{dx} \right)_i = u_x^{[1]}(x_i), \end{cases} \quad (5)$$

where  $u_x^{[1]}(x_i)$  is the first-order derivative of the interpolation function at the cell center. The coefficients of the interpolation function can be given explicitly by the following:

$$\begin{cases} a_3 = \frac{4}{\Delta x_i^3} \left( u_{i+\frac{1}{2}} - u_{i-\frac{1}{2}} - u_x^{[1]}(x_i) \Delta x_i \right), \\ a_2 = \frac{3}{\Delta x_i^2} \left( u_{i-\frac{1}{2}} + u_{i+\frac{1}{2}} - 2\bar{u}_i \right), \\ a_1 = u_x^{[1]}(x_i), \\ a_0 = -\frac{1}{4} \left( u_{i+\frac{1}{2}} + u_{i-\frac{1}{2}} \right) + \frac{3}{2} \bar{u}_i. \end{cases} \quad (6)$$

Here, the first-order derivative  $u_x^{[1]}(x_i)$  can be obtained from an interpolation that involves the VIA and PV from neighboring cells. For example, a fourth-order approximation can be obtained by using the VIAs  $\bar{u}_{i+1}$ ,  $\bar{u}_{i-1}$  and PVs  $u_{i+\frac{1}{2}}$ ,  $u_{i-\frac{1}{2}}$  from two neighboring cells:

$$u_x^{[1]}(x_i) = \frac{10u_{i+\frac{1}{2}} - 10u_{i-\frac{1}{2}} + \bar{u}_{i-1} - \bar{u}_{i+1}}{8\Delta x_i}. \quad (7)$$

With different choices to approximate the first-order derivative  $u_x^{[1]}(x_i)$ , we can modify the interpolation function to reach some desired properties for the resulting numerical scheme. It will be shown later that the  $u_x^{[1]}(x_i)$  can be reconstructed by WENO methodology to yield the essentially non-oscillatory property.

**2.1.2. Time integration.** We have constructed the interpolation function inside each element in the previous subsection. The next problem is how to use the interpolation function to update the solutions  $\bar{u}_i^n$  and  $u_{i+\frac{1}{2}}^n$  from time  $t^n$  to next time step  $t^{n+1} = t^n + \Delta t$  with  $\Delta t$  being the time step increment. We will introduce the time integration procedure of CIP-CSL scheme for moment VIA and PV, respectively, in the following. *Update the PV* For CIP-CSL scheme, we compute the semi-Lagrangian solution  $u_{i+\frac{1}{2}}$  by firstly finding out the departure point for each cell boundary. The departure point can be obtained by solving the initial value problem:

$$\begin{cases} \frac{dX}{dt} = -a(X, t), \\ X(t_0 = t^n) = X_0 = x_{i+\frac{1}{2}}. \end{cases} \quad (8)$$

The ordinary differential equation (8) can be solved by a third-order Runge–Kutta scheme as follows:

$$\begin{cases} X_1 = X_0 - a(X_0, t_0)\Delta t, \\ X_2 = \frac{3}{4}X_0 + \frac{1}{4}X_1 - \frac{1}{4}a(X_1, t_1)\Delta t, \\ X_3 = \frac{1}{3}X_0 + \frac{2}{3}X_2 - \frac{2}{3}a(X_2, t_2)\Delta t, \end{cases} \quad (9)$$

or equivalently

$$\begin{cases} X_1 = X_0 - a(X_0, t_0)\Delta t, \\ X_2 = X_0 - \frac{1}{4}(a(X_0, t_0) + a(X_1, t_1))\Delta t, \\ X_3 = X_0 - \frac{1}{6}(a(X_0, t_0) + a(X_1, t_1) + 4a(X_2, t_2))\Delta t. \end{cases} \quad (10)$$

The governing Equation (1) can be re-written in non-conservative as follows:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} + u \frac{\partial a}{\partial x} = 0, \quad (11)$$

where the second term  $a \frac{\partial u}{\partial x}$  represents the advection part and the third term  $u \frac{\partial a}{\partial x}$  represents the divergence correction part. For advection part, we can compute the semi-Lagrangian solution of PV at each substep of Runge–Kutta integration:

$$u_{i+\frac{1}{2}}^{(l)} = \tilde{u}_i(X_l), \quad l = 1, 2, 3, \quad (12)$$

where  $l$  represents the  $l$ th substep of the third-order Runge–Kutta method.  $\tilde{u}_i(x)$  is the piecewise interpolation function in the form of (4) constructed at  $t = t^n$  over the mesh where the departure point  $X_l$  falls in.

For non-advection part, we can obtain the solution  $u$  by integrating the equation  $\frac{du}{dt} = -u \frac{\partial a}{\partial x}$  along the trajectory. The right-hand side can be approximated by the summation  $\hat{D}_{i+\frac{1}{2}} = -\Delta t \sum_{l'=0}^{l-1} \alpha_{l'} \tilde{u}_i(X_{l'}) \frac{\partial a}{\partial x}(X_{l'})$  where  $\alpha_{l'}$  represents the Runge–Kutta weight coefficients.

Finally, the PV at cell boundary of time  $t = t^{n+1}$  can be given as the sum of the contributions from advection part and non-advection part:

$$u_{i+\frac{1}{2}}^{n+1} = u_{i+\frac{1}{2}}^{(3)} + D_{i+\frac{1}{2}}. \quad (13)$$

*Update the VIA* To update the VIA to the next time step, we adopt an FVM approach by integrating (1) over the interval  $[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ , which yields the following conservative formulation:

$$\bar{u}^{n+1} = \bar{u}^n - \frac{\Delta t}{\Delta x_i} \left( F_{i+\frac{1}{2}}(a, u) - F_{i-\frac{1}{2}}(a, u) \right), \quad (14)$$

where  $F(a, u) = au$  for the transport Equation (1) is the numerical flux that can be computed either by the exact integration of the interpolation function along the trajectory or through a quadrature from the values sampled at the substeps of the Runge–Kutta method. In this work, we use the latter for its simplicity and adequate accuracy. The approximated numerical flux is then given by the following:

$$F_{i+\frac{1}{2}}(a, u) \approx \hat{F}_{i+\frac{1}{2}}(a, u) = \frac{F\left(a(X_0, t^n), u_{i+\frac{1}{2}}^{(0)}\right) + F\left(a(X_1, t^n), u_{i+\frac{1}{2}}^{(1)}\right) + 4F\left(a(X_2, t^n), u_{i+\frac{1}{2}}^{(2)}\right)}{6}, \quad (15)$$

and the VIA is updated by the following:

$$\bar{u}^{n+1} = \bar{u}^n - \frac{\Delta t}{\Delta x_i} \left( \hat{F}_{i+\frac{1}{2}}(a, u) - \hat{F}_{i-\frac{1}{2}}(a, u) \right). \quad (16)$$

## 2.2. Euler conservation laws

The implementation of CIP/MM FVM scheme for Euler conservation laws is based on the hyperbolicity of Euler equations by computing the semi-Lagrangian solution in terms of the characteristic variables. The VIA can be updated through a flux-form formulation, while the numerical flux is approximated from the semi-Lagrangian solution of PV at each Runge–Kutta substep. Here, we also use the total variation diminishing third-order Runge–Kutta time integration method to update the semi-Lagrangian solutions.

**2.2.1. Basic structure of 1D Euler equations.** The conservative form of 1D Euler equations for ideal gas is given by the following:

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0, \quad (17)$$

where

$$\mathbf{U} = \begin{pmatrix} u^{(1)} \\ u^{(2)} \\ u^{(3)} \end{pmatrix} = \begin{pmatrix} \rho \\ \rho v \\ E \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{pmatrix} f^{(1)} \\ f^{(2)} \\ f^{(3)} \end{pmatrix} = \begin{pmatrix} \rho v \\ \rho v^2 + p \\ v(E + p) \end{pmatrix}. \quad (18)$$

Here,  $\rho$  is the density,  $v$  the velocity,  $p$  the pressure, and  $E$  the total energy. We use the equation of state of ideal gas in our calculation, that is,

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho v^2, \quad (19)$$

where  $\gamma = 1.4$ .

The 1D Euler equations can also be formulated in terms of primitive variables in a quasi-linear form

$$\mathbf{W}_t + \mathbf{A}\mathbf{W}_x = 0, \quad (20)$$

where

$$\mathbf{W} = \begin{pmatrix} \rho \\ v \\ p \end{pmatrix}, \quad \mathbf{A} = \begin{bmatrix} v & \rho & 0 \\ 0 & v & 1/\rho \\ 0 & \rho c^2 & v \end{bmatrix}, \quad (21)$$

and  $c = \sqrt{\frac{\gamma p}{\rho}}$  is the sound speed. The Jacobian matrix  $\mathbf{A}$  can be diagonalized by using the left and right eigenvectors

$$\mathbf{A} = \mathbf{R}\mathbf{\Lambda}\mathbf{L}. \quad (22)$$

Here,  $\mathbf{\Lambda}$  is the diagonal matrix of eigenvalues, which are the propagation speeds of the characteristic variables.  $\mathbf{L}$  and  $\mathbf{R}$  denote the matrices of left and right eigenvectors, respectively, which satisfy  $\mathbf{R} = \mathbf{L}^{-1}$ .

Then, Equation (20) can be recast into the characteristic form:

$$\mathbf{L}\mathbf{W}_t + \Lambda\mathbf{L}\mathbf{W}_x = 0, \quad \Lambda = \begin{bmatrix} v & 0 & 0 \\ 0 & v+c & 0 \\ 0 & 0 & v-c \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 1 & 0 & \frac{-1}{c^2} \\ 0 & 1 & \frac{1}{\rho c} \\ 0 & 1 & \frac{-1}{\rho c} \end{bmatrix}. \quad (23)$$

For each characteristic direction  $\frac{dX}{dt} = \lambda_m$ , we have the following characteristic equation:

$$\mathbf{L}^m d\mathbf{W} = 0, \quad m = 1, 2, 3. \quad (24)$$

$\mathbf{L}^m$  is the  $m$ th row vector of matrix  $\mathbf{L}$ . By expanding (24) for each left eigenvector  $\mathbf{L}^1$ ,  $\mathbf{L}^2$ , and  $\mathbf{L}^3$ , we can get the following characteristic equations:

$$d\rho - \frac{1}{c^2}dp = 0 \quad \text{along} \quad \theta_1 : \frac{dX}{dt} = \lambda_1 = v, \quad X(t=0) = X_0, \quad (25a)$$

$$dv + \frac{1}{\rho c}dp = 0 \quad \text{along} \quad \theta_2 : \frac{dX}{dt} = \lambda_2 = v+c, \quad X(t=0) = X_0, \quad (25b)$$

$$dv - \frac{1}{\rho c}dp = 0 \quad \text{along} \quad \theta_3 : \frac{dX}{dt} = \lambda_3 = v-c, \quad X(t=0) = X_0, \quad (25c)$$

where  $\theta_m$ ,  $m = 1, 2, 3$  represents each characteristic curve. Then the primitive variables at  $t = t^{n+1}$  can be found through the following coupled equations:

$$\rho(X_0)^{n+1} - \rho(X(\theta_1)) - \frac{1}{c^2}\{p(X_0)^{n+1} - p(X(\theta_1))\} = 0, \quad (26a)$$

$$v(X_0)^{n+1} - v(X(\theta_2)) + \frac{1}{\rho c}\{p(X_0)^{n+1} - p(X(\theta_2))\} = 0, \quad (26b)$$

$$v(X_0)^{n+1} - v(X(\theta_3)) - \frac{1}{\rho c}\{p(X_0)^{n+1} - p(X(\theta_3))\} = 0, \quad (26c)$$

where  $X(\theta_m)$ ,  $m = 1, 2, 3$  indicates the solution point computed from the interpolation function on each characteristic curve.

**2.2.2. The update of multi-moment for 1D Euler equations.** As mentioned in the previous subsection, we have defined two types of moments inside each cell element, that is,

$$\begin{aligned} \text{PV : } \quad \mathbf{W}_{i+\frac{1}{2}} &= \mathbf{W}\left(x_{i+\frac{1}{2}}, t\right), \\ \text{VIA : } \quad \overline{\mathbf{W}}_i &= \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{W}(x, t) dx. \end{aligned} \quad (27)$$

The departure point can be found by solving the following trajectory equations along the characteristic curves:

$$\begin{cases} \frac{dX}{dt} = -\lambda_m(X, t), \\ X(t=0) = X_0 = x_{i+\frac{1}{2}}, \end{cases} \quad m = 1, 2, 3. \quad (28)$$

Here,  $\lambda_m(X, t)$  is the characteristic speed of each characteristic curve. Same as in the scalar case, we solve the aforementioned ordinary differential equation by the third-order Runge–Kutta method:

$$\begin{cases} X_1(\theta_m) = X_0 - \lambda_m(X_0, t_0)\Delta t, \\ X_2(\theta_m) = \frac{3}{4}X_0 + \frac{1}{4}X_1(\theta_m) - \frac{1}{4}\lambda_m(X_0, t_1)\Delta t, \\ X_3(\theta_m) = \frac{1}{3}X_0 + \frac{2}{3}X_2(\theta_m) - \frac{2}{3}\lambda_m(X_0, t_2)\Delta t, \end{cases} \quad m = 1, 2, 3, \quad (29)$$

where  $\theta_m$  denotes the  $m$ th characteristic curve. By solving the linear system (26), we have the semi-Lagrangian solution of the primitive variables at each Runge–Kutta substep as follows:

$$\begin{aligned} p_{i+\frac{1}{2}}^{(l)} &= \frac{1}{2} \left( p(X_l(\theta_2)) + p(X_l(\theta_3)) + p_{i+\frac{1}{2}}^{(l-1)} c_{i+\frac{1}{2}}^{(l-1)} (v(X_l(\theta_2)) - v(X_l(\theta_3))) \right), \\ v_{i+\frac{1}{2}}^{(l)} &= \frac{1}{2} \left( v(X_l(\theta_2)) + v(X_l(\theta_3)) + \frac{1}{\rho_{i+\frac{1}{2}}^{(l-1)} c_{i+\frac{1}{2}}^{(l-1)}} (v(X_l(\theta_2)) - v(X_l(\theta_3))) \right), \\ \rho_{i+\frac{1}{2}}^{(l)} &= \rho(X_l(\theta_1)) + \frac{1}{\left(c_{i+\frac{1}{2}}^{(l-1)}\right)^2} \left( p_{i+\frac{1}{2}}^{(l)} - p(X_l(\theta_3)) \right), \end{aligned} \quad (30)$$

where  $\rho(x)$ ,  $v(x)$  and  $p(x)$  denote the interpolation functions for the primitive variables  $\rho$ ,  $v$  and  $p$ , respectively.  $\rho(X_l(\theta_m))$ ,  $v(X_l(\theta_m))$  and  $p(X_l(\theta_m))$ ,  $m = 1, 2, 3$ ;  $l = 1, 2, 3$ , are the semi-Lagrangian solutions to the primitive variables along characteristic curve  $\theta_m$  at the  $l$ th Runge–Kutta substep.

Consequently, we can update the PVs of the primitive variables at cell boundary  $x = x_{i+\frac{1}{2}}$  to the next time step  $t^{n+1}$  directly by the following:

$$\begin{aligned} p_{i+\frac{1}{2}}^{n+1} &= p_{i+\frac{1}{2}}^{(3)}, \\ v_{i+\frac{1}{2}}^{n+1} &= v_{i+\frac{1}{2}}^{(3)}, \\ \rho_{i+\frac{1}{2}}^{n+1} &= \rho_{i+\frac{1}{2}}^{(3)}. \end{aligned} \quad (31)$$

The update of the VIA for the conservative variables  $\mathbf{U}$  simply follows the traditional FVM method as follows:

$$\bar{\mathbf{U}}_i^{n+1} = \bar{\mathbf{U}}_i^n - \frac{\Delta t}{\Delta x_i} \left( \hat{\mathbf{F}}_{i+\frac{1}{2}} - \hat{\mathbf{F}}_{i-\frac{1}{2}} \right). \quad (32)$$

Here, the numerical flux  $\hat{\mathbf{F}}_{i+\frac{1}{2}}$  can be obtained by the numerical integration of the PVs from the Runge–Kutta substeps as follows:

$$\hat{\mathbf{F}}_{i+\frac{1}{2}} = \frac{\mathbf{F}(\mathbf{W}_{i+\frac{1}{2}}^{(0)}) + \mathbf{F}(\mathbf{W}_{i+\frac{1}{2}}^{(1)}) + 4\mathbf{F}(\mathbf{W}_{i+\frac{1}{2}}^{(2)})}{6}. \quad (33)$$

### 3. WEIGHTED ESSENTIALLY NON-OSCILLATORY LIMITER

In this section, we will introduce the WENO limiter built from the local information available in the CIP-CSL3 scheme. The basic idea of our WENO limiter is to reconstruct the first-order derivative at the cell center for the interpolation function of CIP-CSL3 scheme. As the local information of



both PV and VIA is used in the WENO interpolation, our WENO limiter is more suitable for multi-moment reconstruction of CIP/MM FVM. Moreover, the limiting projection of the present WENO limiter can be applied to all cells without causing excessive numerical dissipation, unlike other existing WENO limiters that work only for the so-called ‘trouble-cells’ identified by an *ad hoc* TVB criterion.

### 3.1. Weighted essentially non-oscillatory limiter for scalar conservation law

The present WENO reconstruction uses the information of both PV and VIA. For the sake of simplicity, instead of direct use of the VIA, we use the PV at the center of cell  $\Omega_i$  for interpolation, which is computed by the following:

$$\hat{u}_i = \frac{3}{2}\bar{u}_i - \frac{1}{4}\left(u_{i-\frac{1}{2}} + u_{i+\frac{1}{2}}\right). \quad (34)$$

Given the PVs at the cell center  $\hat{u}_i$  and the cell boundaries  $u_{i\pm\frac{1}{2}}$ , we can set the WENO interpolation stencils as shown in Figure 1. Here, we have three small stencils  $S_j = \{x_{i-1+\frac{j}{2}}, x_{i-\frac{j}{2}+\frac{j}{2}}, x_{i+\frac{j}{2}}\}$ ,  $j = 0, 1, 2$ , and one large stencil  $S_3 = \{x_{i-1}, x_{i-\frac{1}{2}}, x_i, x_{i+\frac{1}{2}}, x_{i+1}\}$ . Over each small stencil, we can construct a second-degree polynomial with three PVs. The polynomial obtained from each stencil,  $S_j$ , is denoted by  $q_j(x)$ ,  $j = 0, 1, 2$ . Then, we can compute the first-order derivatives at the cell center with second-order accuracy, respectively, from the derivatives of polynomials  $q_j(x)$ ,  $j = 0, 1, 2$  as follows:

$$\begin{aligned} q_0^{[1]}(x_i) &= \frac{3\hat{u}_i + \hat{u}_{i-1} - 4u_{i-\frac{1}{2}}}{\Delta x_i}, \\ q_1^{[1]}(x_i) &= \frac{u_{i+\frac{1}{2}} - u_{i-\frac{1}{2}}}{\Delta x_i}, \\ q_2^{[1]}(x_i) &= -\frac{3\hat{u}_i + \hat{u}_{i+1} - 4u_{i+\frac{1}{2}}}{\Delta x_i}. \end{aligned} \quad (35)$$

Similarly, we can construct a fourth-degree polynomial in the large stencil  $S_3$ , denoted by  $Q_i(x)$ , and approximate the first-order derivative at the cell center with fourth-order accuracy by the following:

$$Q_i^{[1]}(x_i) = \frac{8u_{i+\frac{1}{2}} + \hat{u}_{i-1} - 8u_{i-\frac{1}{2}} - \hat{u}_{i+1}}{6\Delta x_i}, \quad (36)$$

which can be expressed as a linear combination of the second-order approximations from the three small stencils:

$$Q_i^{[1]}(x_i) = \sum_{j=0}^2 \gamma_j q_j^{[1]}(x_i), \quad (37)$$

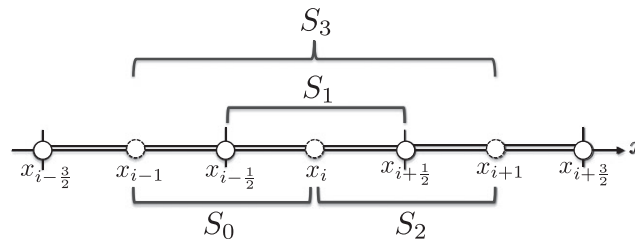


Figure 1. Reconstruction stencils for the weighted essentially non-oscillatory limiter.

where

$$\gamma_0 = \frac{1}{6}, \quad \gamma_1 = \frac{2}{3}, \quad \gamma_2 = \frac{1}{6}. \quad (38)$$

By replacing the linear weights  $\gamma_j$  by nonlinear weights  $\omega_j$  using the WENO methodology, we can achieve the essentially non-oscillatory property for the numerical scheme.

For the WENO reconstruction, we define the smoothness indicator by the following:

$$\beta_j = \sum_{l=1}^2 \int_{x_{i-\frac{1}{8}}}^{x_{i+\frac{1}{8}}} \Delta x_i^{2l-1} \left( \frac{\partial^l q_j(x)}{\partial x^l} \right)^2 dx, \quad j = 0, 1, 2. \quad (39)$$

The smoothness indicator measures the smoothness of the reconstructed polynomials in the target cell  $\Omega_i$ . In order to match the compact WENO reconstruction stencil of the present scheme, we adjust the upper and lower bounds of integration from  $x_{i-\frac{1}{2}}$  and  $x_{i+\frac{1}{2}}$  as used in the conventional WENO scheme to  $x_{i-\frac{1}{8}}$  and  $x_{i+\frac{1}{8}}$  in this work.

In the literature, there are several variants proposed to compute the WENO nonlinear weights following the classical WENO scheme of Jiang and Shu [5], such as WENO-Z scheme [45], WENO-M scheme [46], and the WENO procedure of Shen and Zha [47, 48]. According to the analysis in [45], the classical WENO has accuracy loss in the smooth area near the discontinuity. The WENO-Z and WENO-M schemes are proposed to improve the accuracy of the classic WENO by using different nonlinear weights and smoothness measurements in the reconstruction. Especially, WENO-Z scheme shows good balance between numerical accuracy and algorithmic simplicity. Thus, we adopt the WENO-Z scheme in our WENO reconstruction and calculate the nonlinear weight  $\omega_j$  by the following:

$$\omega_j = \frac{\alpha_j}{\sum_{k=0}^2 \alpha_k}, \quad \alpha_j = \gamma_j \left( 1 + \left( \frac{\tau_5}{\beta_j + \epsilon} \right)^p \right), \quad j = 0, 1, 2, \quad (40)$$

where  $\epsilon$  equals to  $10^{-40}$  and  $\tau_5 = |\beta_2 - \beta_0|$ . The value of  $p$  can be chosen flexibly to optimize the numerical property of the scheme. According to [45], increasing the value of  $p$  can make the scheme closer to the optimal central scheme for smooth function but cause larger numerical dissipation near the discontinuities because of the less weighting of the discontinuous substencil. In our numerical experiments, we use  $p = 1.0$  in the computation unless stated explicitly. Finally, we can get the WENO-reconstructed first-order derivative  $\tilde{Q}^{[1]}(x_i)$  by the following:

$$\tilde{Q}_i^{[1]}(x_i) = \sum_{j=0}^2 \omega_j q_j^{[1]}(x_i). \quad (41)$$

Remarks: As discussed previously, the present WENO reconstruction is implemented in terms of the first-order derivative and only projects the gradient at the cell center, which is substantially different from other existing limiting projections. Shown in our previous works [24, 33, 44, 49], the CIP-CSL3 formulation [24] provides a general framework to design new schemes by modifying the first-order derivative at the cell center. Desired numerical properties, such as oscillationless and diffusionless, can be easily obtained by projecting the first-order derivative with existing approximations, like the WENO in the present paper as the most updated variant. We have explored to use this convenience in other approaches, such as the FVM and the standard DG method, but not reached a conclusion yet. In spite of the success of the CSL3 formulation, limiting projection for high-order local reconstructions still remains an unsolved issue.

It should be also noted that even though the present WENO scheme can effectively suppress numerical oscillations, it does not guarantee the positivity of the solution. Thus, slight under-shoot or over-shoot may be observed in the numerical solutions. We examined the global maximum and minimum values in the numerical results of the following two advection tests.

Table I. Global min/max values in the numerical tests for CIP-CSL-WENO4 scheme.

| Scheme        | Case 1    |                    | Case 2    |                    |
|---------------|-----------|--------------------|-----------|--------------------|
|               | Max value | Min value          | Max value | Min value          |
| CIP-CSL-WENO4 | 1.0       | $-1.74\text{e-}22$ | 1.0       | $-1.48\text{e-}16$ |

Case 1: Advection of a square wave: The initial profile is given by the following:

$$u(x, 0) = \begin{cases} 1, & |x| \leq 0.4, \\ 0, & \text{otherwise,} \end{cases} \quad x \in [-1, 1], \quad (42)$$

over a 100-cell mesh.

Case 2: Advection of a triangle wave [50]: In this test, an initial cone-shaped disturbance with a half width of 10 cells is defined by the following:

$$u(x, 0) = \begin{cases} 1 - 20|x - 0.35|, & 0.3 \leq x \leq 0.4, \\ 0, & \text{otherwise,} \end{cases} \quad x \in [0, 0.7], \quad (43)$$

over a 70-cell mesh.

We carried out computation for one period advection and show the global maximum and minimum values in Table I. It can be seen that there is no overshoot for both cases even with discontinuities, whereas small undershoots are observed.

In some applications, the positivity-preserving property becomes crucial [51], which should be able to be applied to CIP-CSL-WENO4 scheme as well.

As discussed previously, the implementation of the present WENO limiter is efficient because the WENO limiter only needs to compute the first-order derivative at the cell center. To give a quantitative comparison, we measured the computational time of case 1 on a 1000-cell mesh for 10 periods by using the CIP-CSL-WENO4 scheme and the CIP-CSL3 scheme, respectively. The elapse time on a PC with an Intel core-i5 processor (four cores) for CIP-CSL-WENO4 scheme is 7.43 s, while that for CIP-CSL3 scheme is 5.42 s. It shows that the WENO limiter projection causes about a 27% overhead in the computational cost.

### 3.2. Weighted essentially non-oscillatory limiter for 1D Euler equations

For the Euler system, the WENO reconstruction is carried out in terms of the characteristic variables. Firstly, we recover the PV of the primitive variables  $(\rho, v, p)$  at the cell center through Equation (34). Then, the primitive variables  $\mathbf{W} = (\rho, v, p)^T$  at the solution points used for the WENO reconstruction are mapped to the characteristic variables  $\mathbf{C} = (c^{(1)}, c^{(2)}, c^{(3)})^T$  using the left eigenvectors of the Jacobian matrix  $\bar{\mathbf{A}}_i$  computed from the VIA of the target cell  $\Omega_i$  by the following:

$$\mathbf{C}_j = \bar{\mathbf{L}}_i \mathbf{W}_j, \quad j = i - 1, i - \frac{1}{2}, i, i + \frac{1}{2}, i + 1, \quad (44)$$

and  $\bar{\mathbf{L}}_i$  is a  $3 \times 3$  matrix whose rows are the left eigenvectors of Jacobian matrix  $\bar{\mathbf{A}}_i$ . The explicit form of  $\bar{\mathbf{L}}_i$  has been given in (23). The WENO reconstruction procedure introduced in the previous subsection is then applied component-wisely to compute the first-order derivative  $c_i^{(m)[1]}$ ,  $m = 1, 2, 3$ , of the characteristic variables  $\mathbf{C}_i$  at the center of the target cell. The WENO-limited first-order derivatives of the characteristic variables are remapped back to the primitive variables using the following:

$$\mathbf{W}_i^{[1]} = \bar{\mathbf{R}}_i \mathbf{C}_i^{[1]}, \quad (45)$$

where  $\bar{\mathbf{R}}_i$  is a  $3 \times 3$  matrix whose columns are the right eigenvectors of Jacobian matrix  $\bar{\mathbf{A}}_i$ .

Having obtained  $\mathbf{W}_i^{[1]}$ , we can construct the CIP-CSL3 interpolation function for each primitive variable by using the WENO-limited first-order derivatives as the constraint conditions. Finally, the moments are updated using the same procedure as given in Section 2.2.2.

#### 4. COMPARISON BETWEEN EULERIAN AND SEMI-LAGRANGIAN SCHEMES

It is known that the semi-Lagrangian method has some advantages in comparison with the Eulerian method [52, 53], such as less restrictive CFL condition and superior numerical dissipation. Different from the conventional semi-Lagrangian schemes, the present scheme has rigorous numerical conservation. With both the trajectory and numerical flux being evaluated with the third-order Runge–Kutta method, we expect that the present scheme has a numerical accuracy comparable with the Eulerian variant in [44]. To demonstrate this, we compare the total errors, dissipation errors, and dispersion errors between Eulerian MCV-WENO4 scheme [44] and the present CIP-CSL-WENO4 scheme through some benchmark tests for linear advection Equation (1).

The total error  $E_{TOT}$ , dispersion error  $E_{DISP}$ , and dissipation error  $E_{DISS}$  are measured using the method presented in [50]. The total error  $E_{TOT}$  is defined by the mean square error for numerical solutions as follows:

$$E_{TOT} = \frac{1}{N} \sum_{i=0}^N (\bar{u}_i^n - \bar{u}_i^e)^2, \quad (46)$$

where  $\bar{u}_i^n$  is the numerical solution of the VIA for cell  $i$ ,  $\bar{u}_i^e$  represents the exact solution, and  $N$  the mesh number. The total error can be split into the dissipation error  $E_{DISS}$  and dispersion error  $E_{DISP}$ :

$$E_{TOT} = E_{DISS} + E_{DISP}, \quad (47)$$

where

$$\begin{aligned} E_{DISS} &= (\sigma(\bar{u}^n) - \sigma(\bar{u}^e))^2 + (\bar{\bar{u}}^n - \bar{\bar{u}}^e)^2, \\ E_{DISP} &= 2(1 - \rho)\sigma(\bar{u}^n)\sigma(\bar{u}^e). \end{aligned} \quad (48)$$

Here,  $\sigma(\bar{u}^n)$  is the standard deviation of numerical solution  $\bar{u}^n$ ,  $\sigma(\bar{u}^e)$  is the standard deviation of analytic solution  $\bar{u}^e$ , and  $\bar{\bar{u}}^e$  represents the mean value of  $\bar{u}^e$ , while  $\bar{\bar{u}}^n$  is the mean value of  $\bar{u}^n$ .  $\rho$  is the correlation coefficient between  $\bar{u}^n$  and  $\bar{u}^e$  defined by the following:

$$\rho = \frac{\text{cov}(\bar{u}^e, \bar{u}^n)}{\sigma(\bar{u}^n)\sigma(\bar{u}^e)}, \quad (49)$$

where  $\text{cov}(\bar{u}^e, \bar{u}^n)$  is the covariance of  $\bar{u}^n$  and  $\bar{u}^e$ . If  $\bar{u}^e$  and  $\bar{u}^n$  are exactly correlated, that is,  $\rho = 1$ , then the dispersion error defined in (48) will become 0, which means that the numerical error is only from the numerical dissipation.

We compared three error measurements between MCV-WENO4 scheme and CIP-CSL-WENO4 scheme by the following benchmark tests for linear advection equation.

- (1) Advection of a square wave (case 1): This initial profile of this test is given in (42). The three measurements of errors are computed after one period ( $t = 2.0$ ) on a mesh of 100 cells.
- (2) Advection of a triangle wave (case 2): We have introduced the initial structure of this test in (43). The numerical errors are computed up to  $t = 0.7$ .
- (3) Extrema of various smoothness (case 3): This test problem was proposed in [1]. The initial profile is given by the following:

$$u(x + 0.5, 0) = \begin{cases} -x \sin\left(\frac{3}{2}\pi x^2\right), & -1 \leq x \leq -\frac{1}{3}, \\ |\sin(2\pi x)|, & \left|\frac{1}{3}\right| \leq x, \\ 2x - 1 - \sin(3\pi x)/6, & \text{otherwise,} \end{cases} \quad x \in [-1, 1]. \quad (50)$$

The errors  $E_{TOT}$ ,  $E_{DISS}$ , and  $E_{DISP}$  are computed at  $t = 2.0$  on a 100-cell mesh.

- (1) Jiang and Shu's test [5] (case 4): This test for linear advection equation has been widely used to verify the scheme's ability to capture the jump discontinuity while keeping the smooth profile with high accuracy. Initially, we have a profile defined by the following:

$$u(x, 0) = \begin{cases} \frac{1}{6} (G(x, \beta, z - \delta) + G(x, \beta, z + \delta) + 4G(x, \beta, z)), & -0.8 \leq x \leq -0.6, \\ 1, & -0.4 \leq x \leq -0.2, \\ 1 - |10(x - 1)|, & 0.0 \leq x \leq 0.2, \\ \frac{1}{6} (F(x, \alpha, a - \delta) + F(x, \alpha, a + \delta) + 4F(x, \alpha, a)), & 0.4 \leq x \leq 0.6, \\ 0, & \text{otherwise,} \end{cases} \quad (51)$$

where the computation domain is  $[-1, 1]$ . The functions  $F$  and  $G$  are defined by the following:

$$G(x, \beta, z) = \exp(-\beta(x - z)^2), \quad F(x, \alpha, a) = \sqrt{\max(1 - \alpha^2(x - a)^2, 0)}, \quad (52)$$

and the coefficients to determine the initial profile are given by the following:

$$a = 0.5, \quad z = 0.7, \quad \delta = 0.005, \quad \alpha = 10.0, \quad \beta = \log 2 / (36\delta^2). \quad (53)$$

We compute this test for one period ( $t = 2.0$ ) on a 100-cell mesh.

In all these tests, we use the  $CFL = 0.4$  for both MCV-WENO4 scheme and CIP-CSL-WENO4 scheme. We show all the error measurements  $E_{TOT}$ ,  $E_{DISS}$ , and  $E_{DISP}$  of advection tests from cases 1 to 4 in Tables II and III. All the results show that the CIP-CSL-WENO4 scheme of semi-Lagrangian type is more accurate in comparison with the Eulerian MCV-WENO4 scheme.

In order to compare the computational efficiency of the two schemes, we measured the elapse times using the same case as in (42) on a 1000-cell mesh by both CIP-CSL-WENO4 scheme and MCV-WENO4 scheme on a PC with an Intel core-i5 processor of four cores. The elapse time of CIP-CSL-WENO4 scheme for 10-period computation is 7.43s, while that of MCV-WENO4 is 10.63s. It is seen that the load for semi-Lagrangian approach is smaller than Eulerian approach. We may owe this in part to the fact that even though the three-step Runge-Kutta time stepping is used in both schemes, the WENO reconstruction is conducted for only once in CIP-CSL-WENO4 but three times in MCV-WENO4. More importantly, the CIP-CSL-WENO4 can use larger CFL number. Our numerical experiments show that the maximum allowable CFL numbers for the MCV-WENO4 and CIP-CSL-WENO4 are 0.6 and 1.0, respectively. It reveals the significant advantage of CIP-CSL-WENO4 in computational efficiency.

Table II. Comparison of  $E_{TOT}$ ,  $E_{DISS}$ , and  $E_{DISP}$  between MCV-WENO4 scheme and CIP-CSL-WENO4 scheme for cases 1 and 2.

| Scheme        | Case 1    |            |            | Case 2    |            |            |
|---------------|-----------|------------|------------|-----------|------------|------------|
|               | $E_{TOT}$ | $E_{DISS}$ | $E_{DISP}$ | $E_{TOT}$ | $E_{DISS}$ | $E_{DISP}$ |
| MCV-WENO4     | 2.97e-03  | 1.37e-04   | 2.83e-03   | 1.32e-04  | 2.70e-06   | 1.30e-04   |
| CIP-CSL-WENO4 | 1.85e-03  | 6.59e-05   | 1.78e-03   | 3.33e-05  | 5.06e-07   | 3.28e-05   |

Table III. The same in Table II, but for cases 3 and 4.

| Scheme        | Case 3    |            |            | Case 4    |            |            |
|---------------|-----------|------------|------------|-----------|------------|------------|
|               | $E_{TOT}$ | $E_{DISS}$ | $E_{DISP}$ | $E_{TOT}$ | $E_{DISS}$ | $E_{DISP}$ |
| MCV-WENO4     | 8.52e-03  | 9.54e-04   | 7.57e-03   | 3.62e-03  | 4.88e-04   | 3.13e-03   |
| CIP-CSL-WENO4 | 6.23e-03  | 7.51e-04   | 5.48e-03   | 2.14e-03  | 1.78e-04   | 1.95e-03   |

It is noted that even though the present CIP-CSL-WENO4 scheme uses only the neighboring cells as the stencil to build the spatial discretization, which limits the CFL number being less than one for computational stability, a scheme that allows  $\text{CFL} > 1.0$  can be devised straightforwardly in the spirit of semi-Lagrangian scheme. Meanwhile, attention must be paid when the characteristic speeds are variable. In this case, new stability condition has to be built in terms of the so-called Lipschitz number or deformational CFL number [53].

## 5. NUMERICAL EXPERIMENTS

To illustrate the robustness and accuracy of the CIP-CSL-WENO4 method, we compute the numerical results for some widely used benchmark tests in this section.

### 5.1. 1D linear advection equation

The linear advection equation has been given in (1). We set  $a = 1$  in all our numerical examples. The numerical tests with different initial profiles to verify the convergence rate and oscillation-suppressing property of CIP-CSL-WENO4 scheme are given in this subsection. The CFL number equals to 0.8 in all our examples. In some cases, we have tested the numerical scheme on both uniform and nonuniform meshes. The nonuniform meshes are created in the same way as introduced in [41], that is, a random variation  $d$  was specified to the size of each mesh cell by adjusting the cell boundaries as  $x_{i+\frac{1}{2}} + d(r_{i+\frac{1}{2}} - 0.5)\Delta x$ , where  $x_{i+\frac{1}{2}}$  and  $\Delta x$  are the cell boundaries and spacing of the uniform grid.  $r_{i+\frac{1}{2}}$  is a random number of uniform distribution ranging from 0 to 1.

*Accuracy test for 1D advection equation.* In this test problem, we examine the convergence rate of our scheme by using the grid refinement. Initially, the smooth profile is set to be  $u(x, 0) = \sin(\pi x)$ ,  $x \in [-1, 1]$ . The  $L_1$  and  $L_\infty$  errors of VIA at  $t = 2.0$  (after one period) are calculated with a different grid resolution. The mesh number is doubly increased from 20 to 320.

In Table IV, we show the numerical errors and convergence rates of CIP-CSL-WENO4 scheme and Eulerian MCV-WENO4 scheme. Also, we give the numerical results with and without WENO limiter in Table V. It can be clearly observed that our scheme perfectly converges up to fourth-order accuracy, and the use of WENO limiter can maintain both the convergence rate and magnitude of numerical errors. Moreover, the magnitude of both  $L_1$  and  $L_\infty$  errors for CIP-CSL-WENO4 scheme is less than 10% of the MCV-WENO4 scheme that uses a smaller CFL number of 0.4.

Table IV. Numerical errors and convergence rates for 1D advection equation,  $t = 2.0$ .

| $N$ | CIP-CSL-WENO4 scheme |       |            |       | MCV-WENO4 scheme |       |            |       |
|-----|----------------------|-------|------------|-------|------------------|-------|------------|-------|
|     | $L_1$ error          | Order | $L_\infty$ | Order | $L_1$ error      | Order | $L_\infty$ | Order |
| 20  | 4.39e-06             |       | 6.65e-06   |       | 4.95e-05         |       | 7.72e-05   |       |
| 40  | 2.74e-07             | 4.00  | 4.29e-07   | 3.95  | 3.12e-06         | 3.98  | 4.88e-06   | 3.98  |
| 80  | 1.72e-08             | 4.00  | 2.69e-08   | 3.99  | 1.96e-07         | 3.99  | 3.07e-07   | 3.99  |
| 160 | 1.07e-09             | 4.00  | 1.69e-09   | 4.00  | 1.23e-08         | 4.00  | 1.92e-08   | 4.00  |
| 320 | 6.73e-11             | 4.00  | 1.06e-10   | 4.00  | 7.67e-10         | 4.00  | 1.20e-09   | 4.00  |

Table V. Numerical errors and convergence rate for 1D advection equation at  $t = 2.0$ .

| $N$ | CIP-CSL-WENO4 scheme |       |            |       | CIP-CSL3 scheme |       |            |       |
|-----|----------------------|-------|------------|-------|-----------------|-------|------------|-------|
|     | $L_1$ error          | Order | $L_\infty$ | Order | $L_1$ error     | Order | $L_\infty$ | Order |
| 20  | 4.39e-06             |       | 6.65e-06   |       | 4.43e-06        |       | 6.93e-06   |       |
| 40  | 2.74e-07             | 4.00  | 4.29e-07   | 3.95  | 2.74e-07        | 4.02  | 4.31e-07   | 4.01  |
| 80  | 1.72e-08             | 4.00  | 2.69e-08   | 3.99  | 1.72e-08        | 3.99  | 2.69e-08   | 4.00  |
| 160 | 1.07e-09             | 4.00  | 1.69e-09   | 4.00  | 1.07e-09        | 4.00  | 1.69e-09   | 4.00  |
| 320 | 6.73e-11             | 4.00  | 1.06e-10   | 4.00  | 6.73e-11        | 4.00  | 1.06e-10   | 4.00  |

Table VI. Numerical errors and convergence rates of CIP-CSL-WENO4 scheme for 1D advection equation on nonuniform meshes at  $t = 2.0$ .

| $N$ | $L_1$ error | Order of accuracy | $L_\infty$ error | Order of accuracy |
|-----|-------------|-------------------|------------------|-------------------|
| 20  | 4.45e-06    |                   | 6.93e-06         |                   |
| 40  | 2.83e-07    | 4.00              | 4.45e-07         | 3.96              |
| 80  | 1.77e-08    | 4.00              | 2.80e-08         | 3.99              |
| 160 | 1.11e-09    | 3.99              | 1.75e-09         | 4.00              |
| 320 | 6.92e-11    | 4.00              | 1.10e-10         | 4.00              |

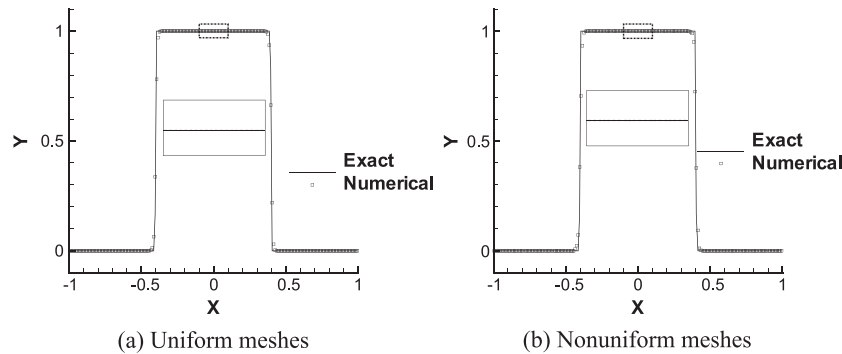


Figure 2. Numerical results of advection of a square wave on a uniform (a) and nonuniform (b) mesh. Part of the region is enlarged to clarify the difference in two meshes.

The results for nonuniform meshes are shown in Table VI. Here, the random variation  $d$  is set to be 0.2. We can see even though the magnitude of numerical error is slightly larger than that on uniform meshes, the convergence rate can be maintained.

*Advection of a square wave.* As the test to examine the capability to capture jump discontinuities, we computed the advection of a square wave. The initial profile is given by the following:

$$u(x, 0) = \begin{cases} 1, & |x - 1| \leq 0.4, \\ 0, & \text{otherwise.} \end{cases} \quad (54)$$

The computational domain is set to be  $[0, 2]$ . The numerical results computed on both uniform and nonuniform meshes of 200 cells at  $t = 2.0$  are shown in Figure 2. The random variation is set to be 0.5 in this example. We can see that our scheme is able to accurately resolve the jump discontinuities.

*Jiang and Shu's test* [5]. The setup of this benchmark test has been defined in (51). The numerical results after one period are given in Figure 3. Here, the mesh number is 200. It can be observed that our numerical solutions are free of spurious oscillations near discontinuities and preserve the high accuracy for smooth profile.

*Extrema of various smoothness.* The initial profile of this test has been given in (50).

We transport the initial irregular profile until  $t = 8.0$  (after four periods) on a 100-cell mesh. Compared with the numerical results in [33], our results in Figure 4 show well-resolved extrema without numerical oscillations near the discontinuities.

## 5.2. 2D implementation

The implementation of semi-Lagrangian schemes in multi-dimensions can be carried out by the Strang splitting for simplicity and computational efficiency using the 1D scheme as the building block. We show the numerical results of transporting a complex distribution by the 2D linear advection equation with a solid-rotating velocity field.

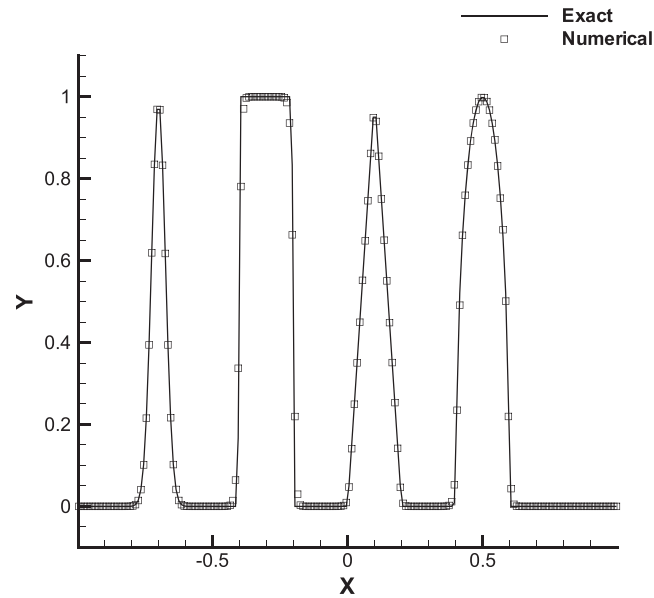


Figure 3. Numerical results of Jiang and Shu's test at  $t = 2.0$  on a 200-cell mesh.

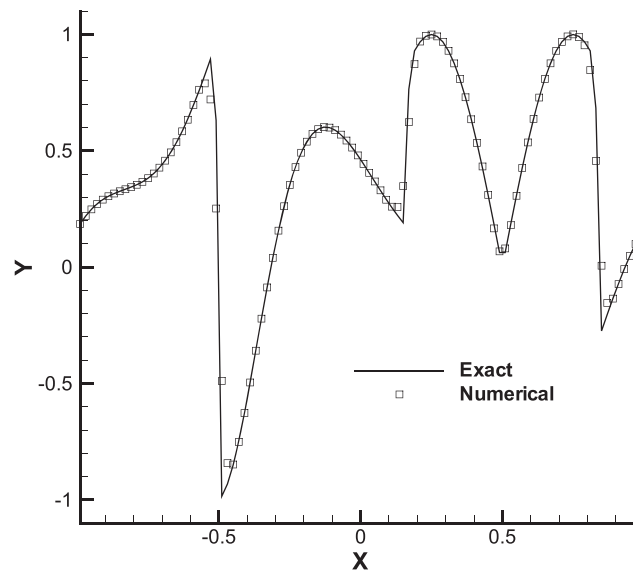


Figure 4. Numerical results of extrema of various smoothness at  $t = 8.0$  on a 100-cell mesh.

In this test, the 2D complex initial profile is given by the following:

$$u(x, y, 0) = \begin{cases} \frac{1}{6} (G(r_1 + \delta, \beta) + G(r_1 - \delta, \beta) + 4G(r_1, \beta)), & |r_1| \leq 0.2, \\ 1, & |x| \leq 0.2, \quad -0.7 \leq y \leq -0.3, \\ 1 - |5r_2|, & |r_2| \leq 0.2, \\ \frac{1}{6} (F(r_3 + \delta, \alpha) + F(r_3 - \delta, \alpha) + 4F(r_3, \alpha)), & |r_3| \leq 0.2, \\ 0, & \text{otherwise,} \end{cases} \quad (55)$$



where

$$r_1 = \sqrt{(x + 0.6)^2 + y^2}, \quad r_2 = \sqrt{(x - 0.6)^2 + y^2}, \quad r_3 = \sqrt{x^2 + (y - 0.6)^2},$$

and  $G(r, \beta) = \exp(-\beta r^2)$ ,  $F(r, \alpha) = \sqrt{\max(1 - \alpha^2 r^2, 0)}$ . As in [21], the coefficients are set to be  $\delta = 0.01$ ,  $\alpha = 5$ , and  $\beta = \log 2 / (36\delta^2)$ . The velocity field is defined by  $(v_1, v_2) = (-2\pi y, 2\pi x)$  on a computation domain of  $[-1, 1]^2$ . Our computation was executed up to  $t = 1.0$  on a grid of  $100 \times 100$  cells. The numerical results are shown in Figure 5.

It can be observed that the shape of the solid body is well resolved, and there are no visible oscillations near the jump discontinuities. The geometric shapes of the distribution have been faithfully preserved as a result of the superior dispersion property of the CIP-CSL-WENO4 scheme.

It should be notified that semi-Lagrangian scheme that relies on the dimensional splitting for multi-dimensional implementation has substantial difficulty for unstructured grids. Moreover, implementing a semi-Lagrangian scheme for the full Euler equations in multi-dimensions meets other complexities, such as the computation of Mach cone. Nevertheless, the strength of semi-Lagrangian method in numerical accuracy and CFL stability makes it attractive in some particular applications.

### 5.3. 1D inviscid Burgers equation

The 1D inviscid Burgers equation is defined as follows:

$$u_t + \left( \frac{u^2}{2} \right)_x = 0, \quad (56)$$

which can be written in an advection form:

$$u_t + a(u)u_x = 0, \quad (57)$$

where  $a(u) = u$  is the characteristic speed. In order to avoid the ‘entropy problem’ at the sonic point, we modify the characteristic speed at the cell boundary  $x_{i+\frac{1}{2}}$  by the following:

$$a(u)_{i+\frac{1}{2}} = \begin{cases} \operatorname{sgn}(\max) \max \left( |u_{i+\frac{1}{2}}|, \left| \frac{1}{2}(u_{i+1} + u_i) \right| \right), & \text{for sonic point,} \\ u_{i+\frac{1}{2}}, & \text{otherwise.} \end{cases} \quad (58)$$

We tested the performance of our scheme to solve Burgers equation for benchmark tests with different initial conditions.

*Burgers equation with a sine wave initial condition.* The initial smooth profile is given by  $u(x, 0) = 0.5 + \sin(\pi x)$ . The solution remains smooth up to  $t = 1.0/\pi$ , and then the shock wave

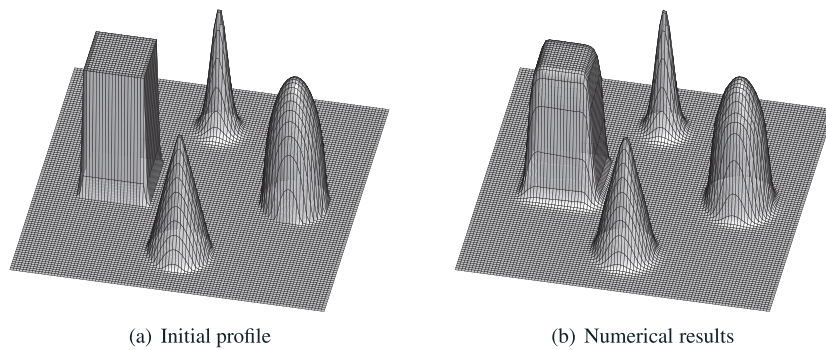
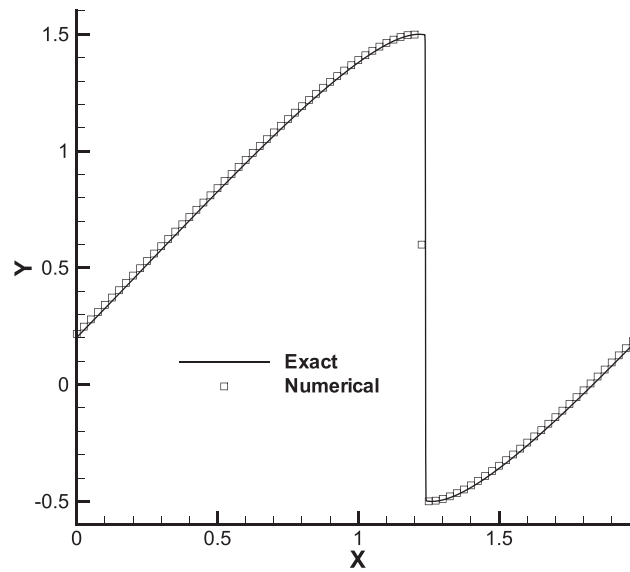


Figure 5. Numerical results of the solid rotation of a complex profile on a mesh of  $100 \times 100$  cells. (a) Initial profile and (b) numerical results.

Table VII. Numerical errors and convergence rates for 1D Burgers equation at  $t = 1/(2\pi)$ .

| $N$ | $L_1$ error | Order of accuracy | $L_\infty$ error | Order of accuracy |
|-----|-------------|-------------------|------------------|-------------------|
| 10  | 3.91e-04    |                   | 2.62e-03         |                   |
| 20  | 4.28e-05    | 3.19              | 2.70e-04         | 3.28              |
| 40  | 3.44e-06    | 3.63              | 3.03e-05         | 3.15              |
| 80  | 2.53e-07    | 3.77              | 2.25e-06         | 3.75              |
| 160 | 2.01e-08    | 3.66              | 2.27e-07         | 3.31              |

Figure 6. Numerical results of Burgers equation at  $t = 1.5/\pi$  on a 100-cell mesh.

starts to develop and becomes a full discontinuity at  $t = 1.5/\pi$ . Firstly, we carried out the computation up to  $t = 0.5/\pi$  with different grid resolutions. The  $L_1$  and  $L_\infty$  errors of VIA are shown in Table VII, along with the convergence rate. It is clear that for 1D Burgers equation, our scheme can achieve fourth-order accuracy as expected. We also give the numerical solutions on a 100-cell mesh at  $t = 1.5/\pi$  in Figure 6. We can see that our scheme can resolve the whole solution with high quality and there is no oscillation near the shock, which verifies the effectiveness of the presented WENO limiter.

*Burgers equation with shock and rarefaction waves.* This benchmark test for 1D inviscid Burgers equation contains both shock and rarefaction waves. The initial profile is a square wave given by the following:

$$u(x, 0) = \begin{cases} 1, & 0.3 \leq x \leq 0.75, \\ 0.5, & \text{otherwise,} \end{cases} \quad (59)$$

on computational domain  $[0, 1]$ . The numerical results at  $t = 0.2$  computed on a mesh of 80 cells are shown in Figure 7. Our numerical solutions can well resolve the shock wave without generating any spurious oscillation, and for the expansion wave, there is no significant dissipation.

#### 5.4. 1D Euler equations

In this section, we report the benchmark tests to verify the performance (convergence rate, stability, and oscillation-suppressing property) of our scheme for 1D Euler equations.

*Accuracy test for 1D Euler equations.* Firstly, we check the convergence rate of the scheme for 1D Euler equations through the propagation of a smooth density perturbation. The initial condition is given by the following:

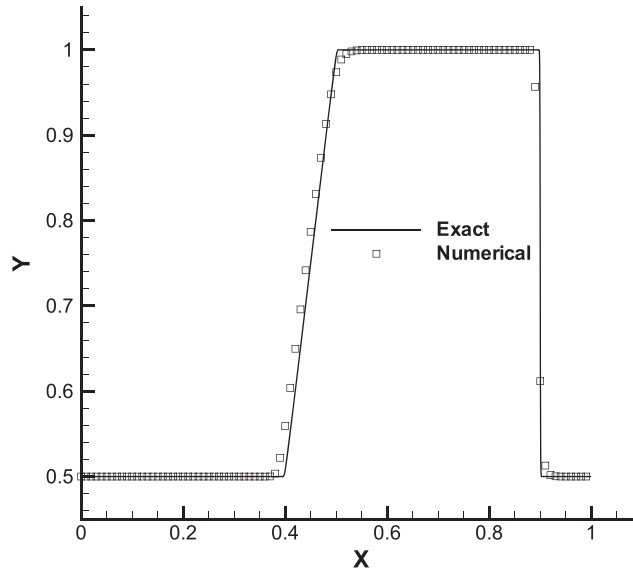


Figure 7. Numerical results of Burgers equation with a shock and rarefaction wave at  $t = 0.2$  on an 80-cell mesh.

Table VIII. Numerical errors and convergence rate for 1D Euler equations,  $t = 2.0$ .

| $N$ | $L_1$ error | Order of accuracy | $L_\infty$ error | Order of accuracy |
|-----|-------------|-------------------|------------------|-------------------|
| 20  | 3.14e-04    |                   | 5.41e-04         |                   |
| 40  | 1.94e-05    | 4.02              | 3.04e-05         | 4.16              |
| 80  | 1.10e-07    | 4.14              | 1.72e-06         | 4.14              |
| 160 | 5.57e-08    | 4.30              | 8.74e-08         | 4.30              |
| 320 | 2.21e-08    | 4.66              | 3.47e-09         | 4.66              |

$$\rho(x, 0) = 1 + 0.2 \sin(\pi x), \quad v(x, 0) = 1.0, \quad p(x, 0) = 1.0, \quad x \in [0, 2]. \quad (60)$$

Here, we impose the periodic boundary condition. We show the  $L_1$  and  $L_\infty$  errors of VIA of density with the grid resolution refined from 20 to 320 in Table VIII. As expected, our scheme is uniformly converged to fourth-order accuracy for the Euler equations with variable characteristic velocities.

*Sod's and Lax's problem.* Here, we show the numerical results for two shock tube problems originally proposed in [3, 54]. For the Sod's problem, the initial distribution is given by the following:

$$(\rho_0, v_0, p_0) = \begin{cases} (1, 0, 1), & 0 \leq x \leq 0.5, \\ (0.125, 0, 0.1), & \text{otherwise.} \end{cases} \quad (61)$$

For the Lax's problem, the initial profile is given by the following:

$$(\rho_0, v_0, p_0) = \begin{cases} (0.445, 0.698, 3.528), & 0 \leq x \leq 0.5, \\ (0.5, 0, 0.571), & \text{otherwise.} \end{cases} \quad (62)$$

For both tests, we use a mesh of 100 cells. In Sod's problem, the computation is carried out up to  $t = 0.25$ . The numerical results shown in Figure 8 reveal that our scheme can capture the contact discontinuity and shock wave with correct position and satisfactory sharpness. In Lax's problem, we conducted the computation until  $t = 0.16$ . Similar to Sod's problem, our numerical results in Figure 9 can resolve the contact discontinuity with high resolution, while the oscillations near the shock wave and discontinuity are effectively eliminated.

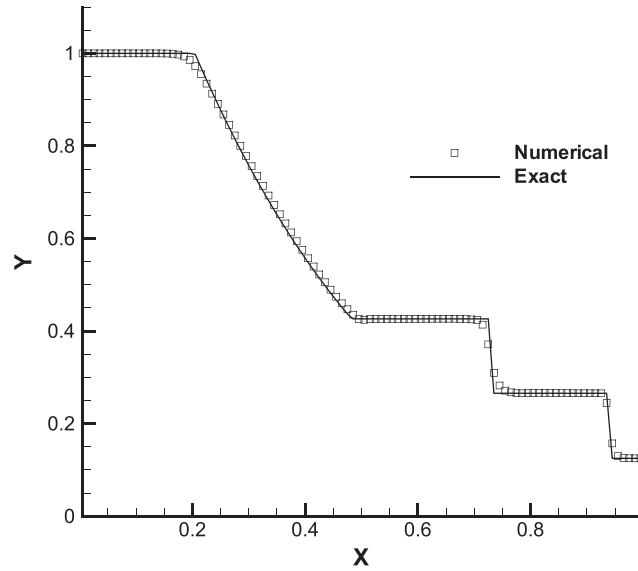


Figure 8. Numerical results of Sod's problem at  $t = 0.25$  with 100 cells.

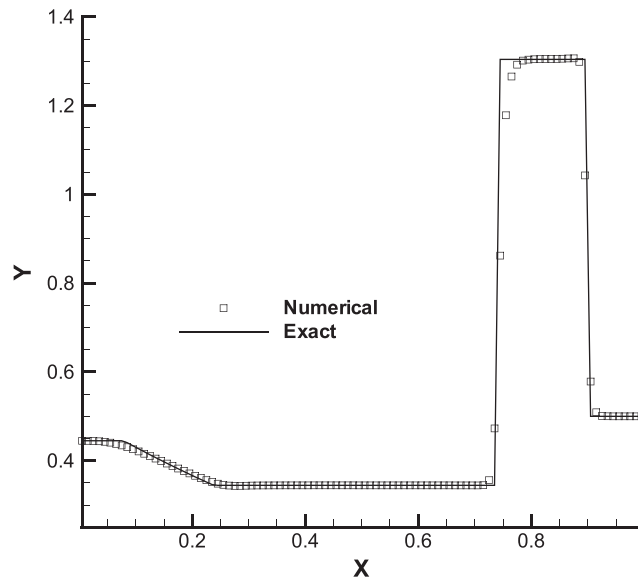


Figure 9. Numerical results of Lax's problem at  $t = 0.16$  with 100 cells.

*Symmetry expansion wave.* This test problem describes an isentropic process where the flow field is split by a divergent velocity field [55]. Initially, we have the following state condition:

$$(\rho_0, v_0, p_0) = \begin{cases} (1, -2, 0.4), & 0 \leq x \leq 0.5, \\ (1, 2, 0.4), & \text{otherwise.} \end{cases} \quad (63)$$

The numerical solutions of density, velocity, pressure, and internal energy at  $t = 0.15$  with 200 cells are shown in Figure 10. For such a test with very low density and pressure, our scheme successfully reproduces all the physical fields with high quality.

*Shock-turbulence interaction.* This benchmark test proposed in [2] includes the interaction of a Mach 3 shock with a density perturbation. The initial profile is given by the following:

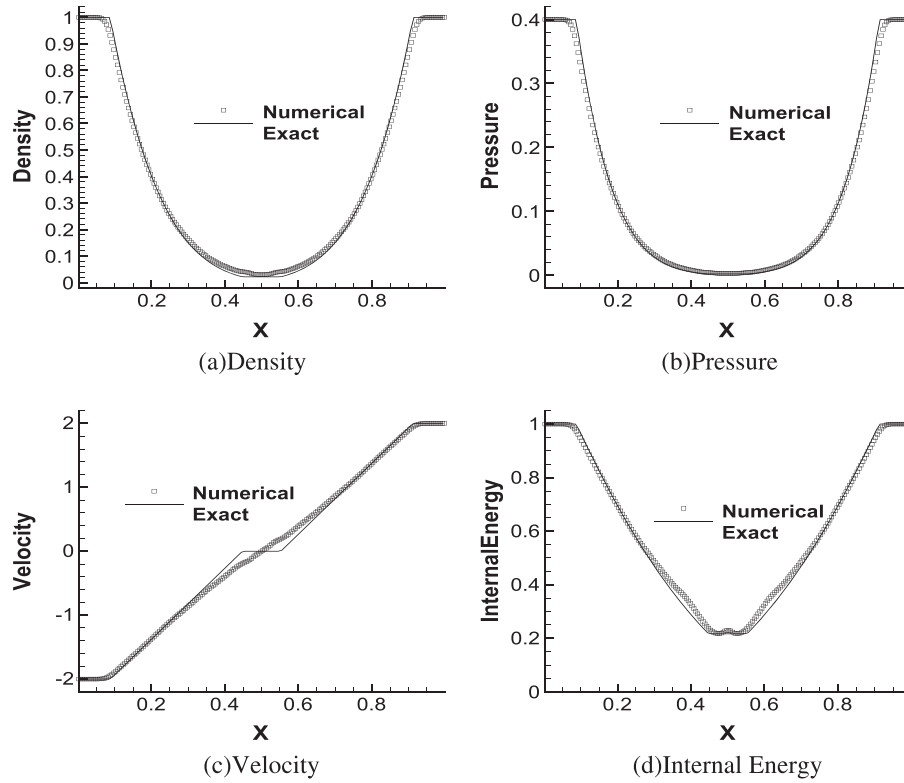


Figure 10. Numerical results of symmetrical expansion wave at  $t = 0.15$  with 200 cells. (a) density, (b) pressure, (c) velocity, and (d) internal energy.

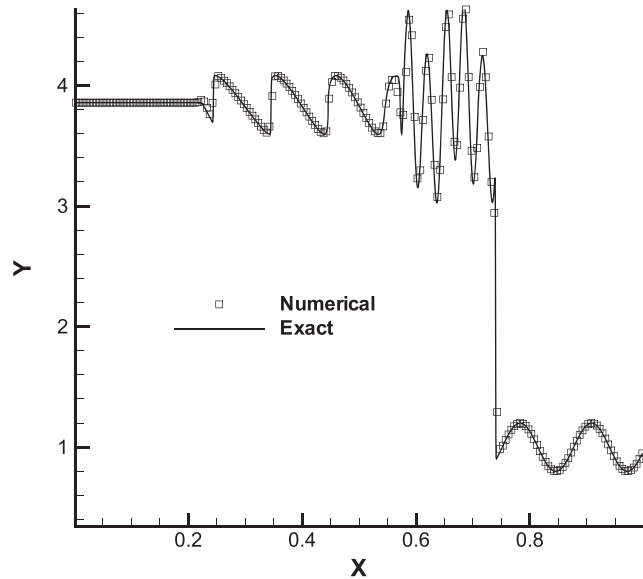


Figure 11. Numerical results of shock-turbulence interaction at  $t = 0.18$  with 200 cells.

$$(\rho_0, v_0, p_0) = \begin{cases} (3.857143, 2.629369, 10.333333), & 0 \leq x \leq 0.1, \\ (1.0 + 0.2 \sin(50x - 25), 0, 1.0), & \text{otherwise.} \end{cases} \quad (64)$$

For this test, a shock wave initially located at  $x = 0.1$  moves right and interacts with the density perturbation superimposed by a sine wave. The numerical results at  $t = 0.18$  are shown in Figure 11.

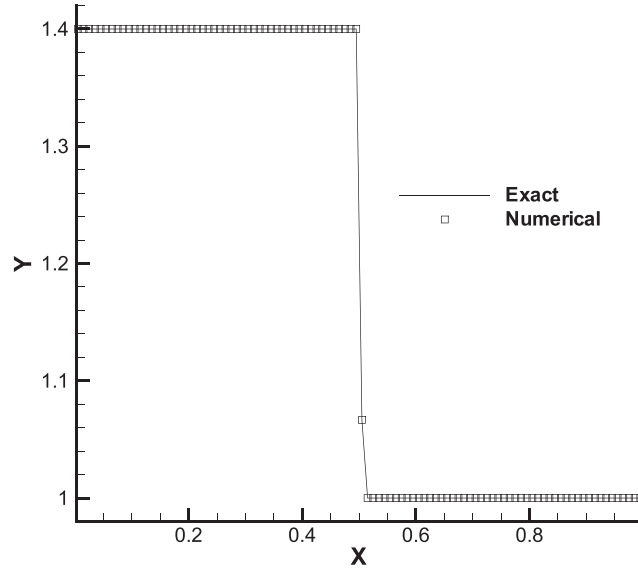


Figure 12. Numerical results of stationary contact discontinuity at  $t = 2.0$  with 100 cells.

The mesh number for this test problem is 200. The density disturbance in the numerical solution has been resolved correctly without obvious diffusion. For the shock wave near  $x = 0.75$ , there are no visible spurious oscillations thanks to the WENO limiter.

*Stationary contact discontinuity.* The initial condition of this test is given by the following:

$$(\rho_0, v_0, p_0) = \begin{cases} (1.4, 0, 1), & 0 \leq x \leq 0.1, \\ (1, 0, 1), & \text{otherwise.} \end{cases} \quad (65)$$

The contact discontinuity defined at the initial condition will remain stationary during the whole computational time. For some high-resolution schemes, it is not a trivial issue to keep the sharpness of the discontinuity for this benchmark test [55]. Here, we carried out the computation up to  $t = 2.0$ . The mesh number is 100. The numerical results are shown in Figure 12. We can see that our scheme is able to keep the density jump with the initial sharpness.

*Two interacting blast waves.* This test was originally proposed by Woodward and Colella in [56]. The initial profile is given by the following:

$$(\rho_0, v_0, p_0) = \begin{cases} (1, 0, 1000), & 0 \leq x \leq 0.1, \\ (1, 0, 0.01), & 0.1 < x < 0.9, \\ (1, 0, 100), & \text{otherwise.} \end{cases} \quad (66)$$

Reflection boundary conditions are imposed on the two ends of the computational domain to generate multiple interactions of different waves, which result in complex structures in the solution.

As in [33], we modified the characteristic velocities at the cell boundary by a weighted averaging between the values of the PVs  $\lambda_m(x_{i+\frac{1}{2}})$  and the values computed by the Roe's average [57] based on the VIAs of two neighboring cells, that is,

$$\hat{\lambda}_m \left( x_{i+\frac{1}{2}} \right) = \beta \lambda_m \left( x_{i+\frac{1}{2}} \right) + (1 - \beta) \bar{\lambda}_m^{Roe} (\bar{u}_i, \bar{u}_{i+1}). \quad (67)$$

We set the weight parameter  $\beta$  to be 0.5 for current example as recommended in [33]. The value of  $p$  in (40) is set to be 1.5 to introduce more numerical dissipations around the strong discontinuities. The numerical results of density at  $t = 0.038$  on a 400-cell mesh are plotted in Figure 13 against the reference solution computed by an FVM with MUSCL reconstruction on a grid of 4000 cells.

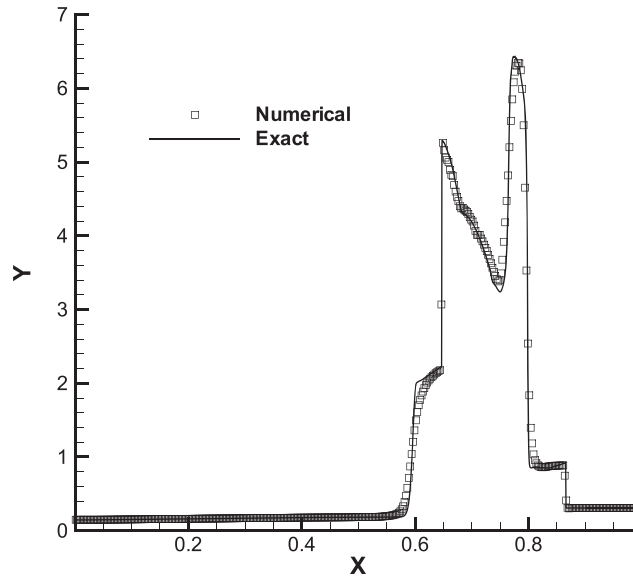


Figure 13. Numerical results of two interacting blast waves at  $t = 0.038$  with 400 cells.

The numerical results shown in Figure 13 accurately capture the shock wave and contact discontinuity with correct positions. The numerical solutions look superior to most existing finite volume WENO schemes on 400-cell mesh and other high-order local reconstruction-based schemes of equivalent DOFs. More importantly, there is no any TVB switching in the present computation to control the numerical dissipation.

## 6. CONCLUSION

In this paper, we present a nodal value-based WENO limiter for CIP/MM FVM scheme. The WENO limiter uses local DOFs available in the target cell and its immediate neighboring cells, which minimizes the WENO interpolation stencil. The present method is substantially different from other existing implementations of WENO projection to local high-order reconstructions that only use the VIA in WENO reconstruction, but ignoring other available sub-cell information, and thus result in wider stencils. The proposed scheme is free of the *ad hoc* TVB criterion that is needed in nearly all existing schemes to control the excessive numerical dissipation. Moreover, the present scheme is algorithmically simple and computationally efficient.

As substantiated in the paper, the semi-Lagrangian formulation is more accurate in terms of both dissipation and dispersion errors in comparison with the Eulerian scheme. Of more practical significance in applications, the presented scheme allows a large CFL number in principle for the computational stability.

The fourth-order convergence rate has been verified for both scalar and Euler conservation laws. Furthermore, the numerical results of some widely used benchmark tests show that the present method is able to get the high-quality solutions among the best ever reported of the numerical schemes with similar number of DOFs.

The present scheme can be seen as a more consistent formulation that well balances different requirements for a high-resolution scheme, such as local high-order reconstruction and limiting projection, numerical accuracy, and computational efficiency.

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